- Compare the pros and cons of solutions obtained using implicit and explicit solvers for static, quasi-static, and dynamic analyses.
- Perform a 1D, 2D, or 3D structural analysis with or without symmetry (axi, cyclic).
- Request desired outputs from commercial FEA software and recall the difference between field and history output data types.
- Setup an FEA analysis to request desired output variables defined spatially and temporally.
- Use commercial FEA software pre-processor to visualize results from an FEA solution.

FEGN527. NONLINEAR APPLICATIONS IN FEA. 3.0 Semester Hrs.
This course explores common nonlinearities frequently encountered in structural applications of FEA. Students will gain proficiency in modeling geometric nonlinearity (large strains), boundary nonlinearity due to contact, and material nonlinearity (creep, rate dependence, plasticity, temperature effects, residual stress). The commercial FEA software Abaqus is used for hands-on experience. Online course. Prerequisite: FEGN526.
Course Learning Outcomes

- Recall and explain the three most common sources of nonlinearity in an FEA simulation.
- Perform an FEA simulation including large strains and finite rotations.
- Execute an FEA simulation including contact and compare several strategies for modeling contact interactions.
- Develop and apply nonlinear models for hyperelastic, viscoelastic, and elastic-plastic materials.
- Use an FEA simulation to compute residual stresses in a part following plastic deformation.
- Construct a clear report to communicate work performed for an FEA simulation.

FEGN528. FEA FOR ADVANCED DESIGN APPLICATIONS. 3.0 Semester Hrs.
In this course students will learn the automation tools and methods necessary for effective application of FEA on advanced design problems. Strategies for parametric analysis, performance optimization, and consideration of statistical uncertainty will be examined using Python scripting and commercial automation software. Online course. Prerequisite: FEGN526.
Course Learning Outcomes

- Apply Python scripting to automate parametric analysis of a part or assembly using commercial FEA software.
- Apply Abaqus Isight to automate parametric analysis of a part or assembly using commercial FEA software.
- Use Python scripting or other software tools to automate extraction and post-processing of results from commercial FEA software.
- Apply automation tools to perform optimization and probabilistic analysis using commercial FEA software.
- Construct a clear report to communicate work performed for an FEA simulation.

FEGN599. INDEPENDENT STUDY. 0.5-6 Semester Hr.

Geochemistry

Program Description

The Graduate Program in Geochemistry is an interdisciplinary program with the mission to educate students whose interests lie at the intersection of the geological, chemical, hydrological and environmental sciences. The Geochemistry Program consists of two subprograms, administering two MS and PhD degree tracks, two professional master's (non-thesis) degree programs, and a graduate certificate. The Geochemistry (GC) degree track pertains to the history and evolution of the Earth and its features, including but not limited to the chemical evolution of the crust and mantle, geochemistry of energy and mineral resources, aqueous geochemistry and fluid-rock/ fluid-mineral interactions and chemical mineralogy. The Environmental Biogeochemistry (EBGC) degree track pertains to the coupled chemical and biological processes of Earth's biosphere, and the changes in these processes caused by human activities.

Primary Contact

Program Director: Jim Ranville
Program Coordinator: Rachel McDonald
geochemistry.mines.edu

Degrees Offered:

- Master of Science/PhD (Geochemistry track)
- Graduate Certificate of Analytical Geochemistry
- Professional Master's Degree in Analytical Geochemistry
- Professional Master's Degree in Environmental Geochemistry

Master of Science and Doctor of Philosophy

1. Geochemistry degree track

Prerequisites

In order to identify any course deficiencies, each entering student will have an entrance interview with their temporary advisor or program director. Credit toward a graduate degree will not be granted for courses taken to fulfill deficiencies.

Program Requirements

The Master of Science (Geochemistry degree track) requires a minimum of 30 credits including:

Course Work	18.0
Research Credits	6.0
Choice or Research or Coursework	6.0
Total Semester Hrs	**30.0**

Table 1

To ensure breadth of background, the course of study for the Master of Science (Geochemistry degree track) must include:		
CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0
CHGC504	METHODS IN GEOCHEMISTRY	3.0
Master of Science (Geochemistry) students select at least 12 credits of the following:		
CHGC509	INTRODUCTION TO AQUEOUS GEOCHEMISTRY	3.0

CHGC514	GEOCHEMISTRY THERMODYNAMICS AND KINETICS	3.0
CEEN550	PRINCIPLES OF ENVIRONMENTAL CHEMISTRY	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
GEGN587	HYDROCHEMICAL AND TRANSPORT PROCESSES	3.0
GEGX571	GEOCHEMICAL EXPLORATION	3.0
GEOL513	HYDROTHERMAL GEOCHEMISTRY	3.0
GEOL523	REFLECTED LIGHT AND ELECTRON MICROSCOPY *	2.0
GEOL540	ISOTOPE GEOCHEMISTRY AND GEOCHRONOLOGY	3.0
Approved Elective Options		
GEGN501	MINERAL DEPOSITS	4.0
GEOL518	MINERAL EXPLORATION	3.0
GEOL535	LITHO ORE FORMING PROCESSES	1.0
Other electives possible with permission of your advisor.		

* Students can add one additional credit of independent study (GEOL599) for XRF methods which is taken concurrently with GEOL523.

Master of Science (Geochemistry degree track) students must also complete an appropriate thesis, based upon original research they have conducted. A thesis proposal and course of study must be approved by the student's thesis committee before the student begins substantial work on the thesis research.

The requirements for the Doctor of Philosophy (Geochemistry degree track) program will be established individually by a student's thesis committee, but must meet the minimum requirements presented below. The Doctor of Philosophy (Geochemistry degree track) program will require a minimum of 72 credits beyond the bachelor's degree. A total of 18 course credits beyond the bachelor's degree are required, with at least 9 credits being completed at Mines. A minimum of 12 research credits are required.

Students who enter the PhD program with a thesis-based master's degree may transfer up to 36 credits in recognition of the coursework and research completed for that degree. At the discretion of the student's thesis committee, up to 24 credits of previous graduate-level coursework (at Mines or elsewhere) can be applied toward the course requirement of the Doctor of Philosophy (Geochemistry degree track) program.

Doctor of Philosophy (Geochemistry degree track) students must take:		
CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0
CHGC504	METHODS IN GEOCHEMISTRY	3.0
Students must also select at least 12 credits of the courses listed in Table 1.		

* Students can add one additional credit of independent study (GEOL599) for XRF methods which is taken concurrently with GEOL523.

Doctor of Philosophy (Geochemistry degree track) students must also complete an appropriate thesis-based upon original research they have conducted. A thesis proposal and course of study must be approved by the student's thesis committee before the student begins substantial work on the thesis research.

Master of Science (Geochemistry degree track) will be expected to give one public seminar on their research and Doctor of Philosophy (Geochemistry degree track) students are required to give at least one public seminar in addition to their thesis defense presentation.

2. Environmental Biogeochemistry (EBGC) Degree track

Prerequisites

A candidate for an MS or PhD in the EBGC degree track should have an undergraduate science or engineering degree with coursework including multivariable calculus, two semesters each of physics and chemistry, and one semester each of biology and earth science. Applicants who do not fulfill these requirements may still be admitted but will need to undergo an entrance interview to establish deficiency requirements. Credit toward a graduate degree will not be given for undergraduate courses taken to fulfill deficiencies.

Program Requirements

Required Curriculum: A thesis proposal and thesis are required for all MS and PhD degrees in the EBGC degree track. MS thesis advisors (or at least one co-advisor) must be members of the EBGC subprogram. PhD thesis committees must have a total of at least four members. PhD advisors (or at least one of two co-advisors) and one additional committee member must be members of the EBGC subprogram. MS students will be expected to give one public seminar on their research; PhD students are required to give at least one in addition to their thesis defense presentation.

In addition, both MS and PhD students in the EBGC degree track must complete the following coursework:		
1. Two Required Classes:		
CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0
CHGC504	METHODS IN GEOCHEMISTRY	3.0
2. One Chemistry-Focused Class, chosen from the following list:		
CHGC505	INTRODUCTION TO ENVIRONMENTAL CHEMISTRY	3.0
CHGC509	INTRODUCTION TO AQUEOUS GEOCHEMISTRY	3.0
CEEN550	PRINCIPLES OF ENVIRONMENTAL CHEMISTRY	3.0
CEEN551	ENVIRONMENTAL ORGANIC CHEMISTRY	3.0
3. One Biology-Focused Class, chosen from the following list:		
CEEN560	MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT	3.0
CEEN562	ENVIRONMENTAL GEOMICROBIOLOGY	3.0
4. One Earth Science-Focused class, chosen from the following list:		
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGX571	GEOCHEMICAL EXPLORATION	3.0

GEOL540	ISOTOPE GEOCHEMISTRY AND GEOCHRONOLOGY	3.0
5. At least One elective approved by your advisor.		
Total Credits Required for MS:		**30.0**
Total Credits Required for PhD:		**72.0**

The student's thesis committee may specify additional course requirements and makes final decisions regarding transfer credits.

Students who enter the PhD program with a thesis-based master's degree may transfer up to 36 credits in recognition of the coursework and research completed for that degree. At the discretion of the student's thesis committee, up to 24 credits of previous graduate-level coursework (at Mines or elsewhere) can be applied toward the course requirement of the Doctor of Philosophy (Geochemistry degree track) program.

A total of 18 course credits are required with at least 9 credits being completed at Mines.

Comprehensive Examination

Doctor of Philosophy (Geochemistry) students in both degree tracks must take a comprehensive examination. It is expected that this exam will be completed within three years of matriculation or after the bulk of coursework is finished, whichever occurs earlier. This examination will be administered by the student's thesis committee and will consist of an oral and a written examination, administered in a format to be determined by the thesis committee. Two negative votes in the thesis committee constitute failure of the examination.

In case of failure of the comprehensive examination, a re-examination may be given upon the recommendation of the thesis committee and approval of the Dean of Graduate Studies. Only one re-examination may be given.

Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Graduate Certificate of Analytical Geochemistry

This program offers an opportunity for working professionals to complete graduate level coursework leading to a Graduate Certificate in a short time. The program focuses on providing instruction in a.) the fundamentals of geochemical analysis, which give students the flexibility to respond to changing opportunities in earth and environmental science, and b.) the mechanics of common techniques (sample collection and preparation, XRF, SEM, and ICP), which prepare students for immediate entry into the jobs market. The Certificate program is comprised of 9.0 credits of coursework. Please note that courses cannot be double counted to fulfill the requirements of other Certificates.

Students are required to take at least 6.0 credits out of the following core courses:

Core Courses:		
CHGC504	METHODS IN GEOCHEMISTRY	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEOL523	REFLECTED LIGHT AND ELECTRON MICROSCOPY *	2.0
GEOL540	ISOTOPE GEOCHEMISTRY AND GEOCHRONOLOGY	3.0

Students can choose up to 3.0 credits out of the following elective courses:

Electives:		
CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0
CEEN560	MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT	3.0
CEEN562	ENVIRONMENTAL GEOMICROBIOLOGY	3.0
CHGN583	PRINCIPLES AND APPLICATIONS OF SURFACE ANALYSIS TECHNIQUES	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
GEGN587	HYDROCHEMICAL AND TRANSPORT PROCESSES	3.0
GEGX571	GEOCHEMICAL EXPLORATION	3.0
GEOL513	HYDROTHERMAL GEOCHEMISTRY	3.0
GEOL628	ADVANCED IGNEOUS PETROLOGY	3.0
MTGN605	ADVANCED TRANSMISSION ELECTRON MICROSCOPY	2.0
PHGN504	RADIATION DETECTION AND MEASUREMENT	3.0

* Students can add one additional credit of independent study (GEOL599) for XRF methods which is taken concurrently with GEOL523.

Professional Masters' Degree in Analytical Geochemistry

This program is designed for both full time students and working professionals who want to increase their knowledge and skills, while gaining a thorough up-date of advances across the spectrum of geochemical analysis.

A minimum of 30 credits are required for the PM degree. Students working towards a PM in Analytical Geochemistry are required to take at least 9.0 credits from courses in the Certificate in Analytical Geochemistry core and 3.0 credits from the electives. Up to 6 credits of the additional electives can include an independent study project (CHGC 598) at Mines, federal agencies, or industry, and is highly encouraged.

The Certificate in Analytical Geochemistry can be combined with the Certificate in Exploration Methods to be used towards the Professional Masters in Analytical Geochemistry (Non-thesis), which will allow part-time students to stack their education. When stacking the Certificates, the additional 6 credits required to complete the PM can include an independent study project (CHGC 599) at Mines, federal agencies, or industry, and is highly encouraged. Independent study projects that connect students with local entities will be a priority and will establish a network of future employers for the students and collaborations for CSM.

Professional Master's Degree in Environmental Geochemistry

The Professional Master's in Environmental Geochemistry program is intended to provide:

1. an opportunity for Mines undergraduates to obtain, as part of a fifth year of study, a Master's in addition to the Bachelor's degree; and
2. additional education for working professionals in the area of geochemistry as it applies to problems relating to the environment.

This is a non-thesis Master's degree program administered by the Environmental Biogeochemistry subprogram of the Geochemistry program, and may be completed as part of a Combined degree program by individuals already matriculated as undergraduate students at Mines, or by individuals already holding undergraduate or advanced degrees and who are interested in a graduate program that does not have the traditional research requirement. The program consists primarily of coursework in geochemistry and allied fields with an emphasis on environmental applications. No research is required though the program does allow for independent study, professional development, internship, and cooperative experience.

A 9 Credit Core Program Consists of:

CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0
CHGC509	INTRODUCTION TO AQUEOUS GEOCHEMISTRY	3.0
or CEEN550	PRINCIPLES OF ENVIRONMENTAL CHEMISTRY	
GEGN566	GROUNDWATER ENGINEERING	3.0
Total Semester Hrs		**9.0**

In addition, 15 credits must be selected from the list below, representing the following core areas: geochemical methods, geographic information system, geological data analysis, groundwater engineering or modeling, hydrothermal geochemistry, isotope geochemistry, physical chemistry, microbiology, mineralogy, organic geochemistry, and thermodynamics.

CHGC504	METHODS IN GEOCHEMISTRY	3.0
CEEN560	MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT	3.0
CEEN562	ENVIRONMENTAL GEOMICROBIOLOGY	3.0
CHGC555	ENVIRONMENTAL ORGANIC CHEMISTRY	3.0
CHGN503	ADV PHYSICAL CHEMISTRY I	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN581	ANALYTICAL HYDROLOGY	3.0
GEGN583	MATHEMATICAL MODELING OF GROUNDWATER SYSTEMS	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
GEGX571	GEOCHEMICAL EXPLORATION	3.0
GEOL540	ISOTOPE GEOCHEMISTRY AND GEOCHRONOLOGY	3.0

An additional 6 credits of free electives may be selected to complete the 30 credit requirement. Free electives may be selected from the course offerings of the Department of Geology and Geological Engineering, the Department of Chemistry, or the Department of Civil and Environmental Engineering, and may also be independent study credits taken to fulfill a research cooperative, or other professional development experience. A course program will be designed in advanced through consultation between the student and an advisor from the Geochemistry Committee of the Whole

Courses

CHGC503. INTRODUCTION TO GEOCHEMISTRY. 3.0 Semester Hrs.
(I) A comprehensive introduction to the basic concepts and principles of geochemistry, coupled with a thorough overview of the related principles of thermodynamics. Topics covered include: nucleosynthesis, origin of earth and solar system, chemical bonding, mineral chemistry, elemental distributions and geochemical cycles, chemical equilibrium and kinetics, isotope systematics, and organic and biogeochemistry. Prerequisite: Introductory chemistry, mineralogy and petrology. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Demonstrate grasp on core concepts of Geochemistry

CHGC504. METHODS IN GEOCHEMISTRY. 3.0 Semester Hrs.
(II) Field sampling of natural earth materials including rocks, soils, sediments, and waters. Preparation of naturally heterogeneous materials, digestions, and partial chemical extractions. Principles of instrumental analysis including trace elemental analysis by ICP-atomic spectroscopy, isotope analysis by ICP-MS, EM/X-ray methods, and chromatography. Quality assurance and quality control. Interpretation and assessment of geochemical data using statistical methods. Course format is hands-on, project oriented. Prerequisite: Graduate standing in geochemistry or environmental science and engineering. 2 hours lecture, 3 hours lab; 3 semester hours.
Course Learning Outcomes

- Graduates will contribute to the advancement of their chosen fields through interpretation and assessment of statistical methods of geochemical data.

CHGC505. INTRODUCTION TO ENVIRONMENTAL CHEMISTRY. 3.0 Semester Hrs.
Equivalent with CHGN403,
(II) Processes by which natural and anthropogenic chemicals interact, react, and are transformed and redistributed in various environmental compartments. Air, soil, and aqueous (fresh and saline surface and groundwaters) environments are covered, along with specialized environments such as waste treatment facilities and the upper atmosphere. Meets with CHGN403. CHGN403 and CHGC505 may not both be taken for credit.

CHGC506. WATER ANALYSIS LABORATORY. 2.0 Semester Hrs.
Instrumental analysis of water samples using spectroscopy and chromatography. Methods for field collection of water samples and field measurements. The development of laboratory skills for the use of ICP-AES, HPLC, ion chromatography, and GC. Laboratory techniques focus on standard methods for the measurement of inorganic and organic constituents in water samples. Methods of data analysis are also presented. Prerequisite: Introductory chemistry, graduate standing. 3 hour laboratory, 1 hour lecture, 2 semester hours.

CHGC508. ANALYTICAL GEOCHEMISTRY. 3.0 Semester Hrs.
Sample preparation and instrumental analysis of geochemical materials using ICP-AES, ICP-MS, LA-ICP-MS, XRF and XRD are key activities in the field of earth resources and environmental monitoring. Course format is hands-on, project oriented. Laboratory techniques focus on industry-

standard methods for the measurement of inorganic constituents, QA/QC, instrument trouble-shooting, and practical exercises with field collected samples. Methods of data interpretation are also key aspects of the course.

Course Learning Outcomes

- At the completion of the course students will:

CHGC509. INTRODUCTION TO AQUEOUS GEOCHEMISTRY. 3.0 Semester Hrs.
Analytical, graphical and interpretive methods applied to aqueous systems. Thermodynamic properties of water and aqueous solutions. Calculations and graphical expression of acid-base, redox and solution-mineral equilibria. Effect of temperature and kinetics on natural aqueous systems. Adsorption and ion exchange equilibria between clays and oxide phases. Behavior of trace elements and complexation in aqueous systems. Application of organic geochemistry to natural aqueous systems. Light stable and unstable isotopic studies applied to aqueous systems. Prerequisite: CHGN209 or CBEN210.

CHGC511. GEOCHEMISTRY OF IGNEOUS ROCKS. 3.0 Semester Hrs.
A survey of the geochemical characteristics of the various types of igneous rock suites. Application of major element, trace element, and isotope geochemistry to problems of their origin and modification. Prerequisite: Undergraduate mineralogy and petrology. 3 hours lecture; 3 semester hours. Offered alternate years.

CHGC514. GEOCHEMISTRY THERMODYNAMICS AND KINETICS. 3.0 Semester Hrs.
(II) Fundamental principles of classical thermodynamics and kinetics with specific application to the earth sciences. Volume-temperature – pressure relationships for solids, liquids, gases and solutions. Energy and the First Law, Entropy and the Second and Third Laws. Gibbs Free Energy, chemical equilibria and the equilibrium constant. Solutions and activity-composition relationships for solids, fluids and gases. Phase equilibria and the graphical representation of equilibira. Application of the fundamentals of kinetics to geochemical examples. Prerequisite: Introductory chemistry, introductory thermodynamics, mineralogy and petrology. 3 hours lecture, 3 semester hours. Offered in alternate years.

CHGC527. ORGANIC GEOCHEMISTRY OF FOSSIL FUELS AND ORE DEPOSITS. 3.0 Semester Hrs.
A study of organic carbonaceous materials in relation to the genesis and modification of fossil fuel and ore deposits. The biological origin of the organic matter will be discussed with emphasis on contributions of microorganisms to the nature of these deposits. Biochemical and thermal changes which convert the organic compounds into petroleum, oil shale, tar sand, coal and other carbonaceous matter will be studied. Principal analytical techniques used for the characterization of organic matter in the geosphere and for evaluation of oil and gas source potential will be discussed. Laboratory exercises will emphasize source rock evaluation, and oil-source rock and oil-oil correlation methods. Prerequisite: CHGN221, GEGN438. 2 hours lecture; 3 hours lab; 3 semester hours. Offered alternate years.

CHGC555. ENVIRONMENTAL ORGANIC CHEMISTRY. 3.0 Semester Hrs.
A study of the chemical and physical interactions which determine the fate, transport and interactions of organic chemicals in aquatic systems, with emphasis on chemical transformations of anthropogenic organic contaminants. Offered in alternate years. 3 hours lecture; 3 semester hours.

CHGC563. ENVIRONMENTAL MICROBIOLOGY. 2.0 Semester Hrs.
An introduction to the microorganisms of major geochemical importance, as well as those of primary importance in water pollution and waste treatment. Microbes and sedimentation, microbial leaching of metals from ores, acid mine water pollution, and the microbial ecology of marine and freshwater habitats are covered. Prerequisite: none. 1 hour lecture, 3 hours lab; 2 semester hours. Offered alternate years.

CHGC564. BIOGEOCHEMISTRY AND GEOMICROBIOLOGY. 3.0 Semester Hrs.
Designed to give the student an understanding of the role of living things, particularly microorganisms, in the shaping of the earth. Among the subjects will be the aspects of living processes, chemical composition and characteristics of biological material, origin of life, role of microorganisms in weathering of rocks and the early diagenesis of sediments, and the origin of petroleum, oil shale, and coal. Prerequisite: none. 3 hours lecture; 3 semester hours.

CHGC598. SPECIAL TOPICS. 1-6 Semester Hr.
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

CHGC599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

CHGC599. INDEPENDENT STUDY. 1-6 Semester Hr.

CHGC599. INDEPENDENT STUDY. 1-6 Semester Hr.

CHGC698. SPECIAL TOPICS. 1-6 Semester Hr.
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

CHGC699. INDEPENDENT STUDY. 0.5-6 Semester Hr.
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

CHGC699. INDEPENDENT STUDY. 0.5-6 Semester Hr.

CHGC699. INDEPENDENT STUDY. 0.5-6 Semester Hr.

CHGC699. INDEPENDENT STUDY. 0.5-6 Semester Hr.

CHGC707. GRADUATE THESIS / DISSERTATION RESEARCH CREDIT. 0.5-15 Semester Hr.

Professors

Alexis Navarre-Sitchler, Geology and Geological Engineering

Bettina M. Voelker, Chemistry

Christopher P. Higgins, Civil and Environmental Engineering

James F. Ranville, Chemistry

John McCray, Civil and Environmental Engineering

John R. Spear, Civil and Environmental Engineering

Jonathan Sharp, Civil and Environmental Engineering

Linda A. Figueroa, Civil and Environmental Engineering

Thomas Monecke, Geology and Geological Engineering

Zhaoshan Chang, Geology and Geological Engineering

Associate Professors

Mathias Burisch, Geology and Geological Engineering

Yvette Kuiper, Geology and Geological Engineering

Assistant Professor

Ryan Venturelli, Geology and Geological Engineering

Affiliate Faculty

Katharina Pfaff, Geology and Geological Engineering, Research Associate Professor

Emeriti Professor

John B. Curtis, Geology and Geological Engineering

Wendy J. Harrison, Geology and Geological Engineering

Donald L. Macalady, Chemistry

Patrick MacCarthy, Chemistry

Samuel B. Romberger, Geology and Geological Engineering

Richard F. Wendlandt, Geology and Geological Engineering

Thomas R. Wildeman, Chemistry

Emeriti Associate Professor

L. Graham Closs, Geology and Geological Engineering

E. Craig Simmons, Chemistry

GIS & GeoInformatics

Program Description

The interdisciplinary online program in Geographic Information Systems and GeoInformatics (GIS & GeoInformatics) focuses on the applications of geospatial technology. It provides hands-on geospatial training, multi-criteria decision-making, advanced applications, and quantitative analysis in GIS and Remote Sensing (RS). This program aligns directly with the Colorado School of Mines' strengths in Earth, Energy, and Environment. Our programs enhance students' quantitative geospatial analytical skills, position them ahead of the technology curve, and empower professionals to advance their careers.

Student Learning Outcomes

Students are expected to know and be able to do the following by the time of graduation.

- Master the fundamental concepts of GIS and the skills necessary for conducting geospatial analyses using GIS and/or RS.
- Master the essential principles of optical remote sensing and apply them in image processing.
- Learn and apply the key concepts and principles of Synthetic Aperture Radar (SAR) and Interferometric SAR (InSAR) in InSAR processing.
- Conduct digital elevation model (DEM) acquisition and preparation, creating new map products on a GIS platform to detect spatial and temporal variations of phenomena, such as geomorphic features, landslides, flooding, or forest fires.
- Identify topographic features through the superimposed Digital Elevation Model (DEM) and DEM-derived map products.
- Develop custom Python-based geoprocessing tools and enhance the existing capabilities of GIS applications through Python scripting.
- Automate geoprocessing tasks and#complex spatial#workflows via Python scripting.
- Design and conduct GIS-based projects integrating various datasets, geospatial knowledge, and analytical skills.

Primary Contact

Program Director: Wendy Zhou
Program Coordinator: Rachel McDonald

https://gis.mines.edu/

Certificate and Degree Requirements

We offer a graduate certificate and a non-thesis Master of Science degree. The courses taken for the certificate degree can be used towards the Master's degree. These programs are available as an online program.

The Master of Science (Non-Thesis) Program in GIS & GeoInformatics

The Master of Science (Non-Thesis) Program outlined below may be completed by individuals already holding an undergraduate or advanced degree or as a combined degree program by individuals already matriculated as undergraduate students at Colorado School of Mines. Courses taken while working on any of the four GIS & GeoInformatics graduate certificates can be applied to this Master of Science program. The program is comprised of a total of 30 credit hours.

All Master of Science (Non-Thesis) program will include the following core requirements:

GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN590	GIS-BASED REAL WORLD LEARNING PROJECT I - FUNDAMENTALS	3.0
GEGN592	GIS-BASED REAL WORLD LEARNING PROJECT II - ADVANCED APPLICATIONS	3.0

And, seven of the following courses:

GEGN542	ADVANCED DIGITAL TERRAIN ANALYSIS	3.0
GEGN579	PYTHON SCRIPTING FOR GEOGRAPHIC INFORMATION SYSTEMS	3.0